

SAI VARUN CHETRYPALLY

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SUMMARY

Master's student in Computing and Information Systems seeking student researcher opportunities, with hands-on experience working with **Python**, **Machine Learning**, and **software engineering projects**.

EDUCATION

Master in Computing and Information Systems | Youngstown State University | GPA 3.8 Expected Fall 2026

Bachelor in Electronics and Communications Engineering | MVSR Engineering College | GPA 3.5 2020- 2024

PROFESSIONAL EXPERIENCE

Data Systems Intern – Defense Research & Development Organization, Hyd, India Summer 2023

- Built and supported a database-backed inventory management system for users to store, search, update, and manage structured operational records.
- Developed MySQL tables, backend query logic, and SQL queries to support accurate data entry, retrieval, reporting, validation, and troubleshooting.
- Worked with React.js, Node.js/PHP, HTML, CSS, and database connections to support internal application functionality and improve usability.
- Collaborated with technical team members and documented workflows, setup steps, database logic, and troubleshooting notes to support maintainability.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, SQL
- **AI & Machine Learning:** Machine Learning, Deep Learning, TensorFlow, scikit-learn, CNNs, Model Training, Model Evaluation
- **NLP & AI Systems:** LangChain, OpenAI API, ChromaDB, Embeddings, RAG, Semantic Search
- **Data Science:** Pandas, NumPy, Matplotlib, Data Cleaning, Feature Engineering, Data Analysis, Data Visualization
- **Software Engineering:** FastAPI, Flask, REST APIs, Object-Oriented Programming, Data Structures, Algorithms
- **Databases & Tools:** MySQL, SQLite, Oracle SQL, Git, GitHub, VS Code, Jupyter Notebook, Anaconda

PROJECTS

DocSense: AI-Powered Document Assistant

- Developed a retrieval-based AI document assistant that allows users to upload PDFs, process text, generate embeddings, and retrieve relevant answers through semantic search.
- Built FastAPI endpoints, document chunking, vector search, retrieval workflows, and source-based response logic while testing retrieval accuracy and documenting system behavior.

Pneumonia Detection Using EfficientNetB0

- Developed a deep learning image classification system to detect pneumonia from chest X-ray images using EfficientNetB0, preprocessing, model prediction, and evaluation workflows.
- Published a research paper based on this project in the International Journal of Research and Analytical Reviews (IJRAR), Volume 11, Issue 2, June 2024, Paper ID: IJRAR24B3948.

AI-Assisted Health Monitoring System

- Developed a health monitoring application to process, store, validate, and analyze structured physiological data such as heart rate, SpO₂, respiratory rate, and temperature.
- Built anomaly detection, visualization, reporting, testing, and documentation workflows to support data quality, monitoring, and reliable application behavior.

AI-Powered Code Analyzer and Mini Compiler

- Designed and implemented a code analysis system supporting lexical analysis, parsing, semantic validation, intermediate execution, and structured code review logic.
- Applied object-oriented programming, data structures, algorithmic logic, debugging, and testing to detect syntax issues, validate code behavior, and analyze program complexity.

CERTIFICATIONS

AWS Academy Cloud Foundations – AWS

Python for Data Science – IBM

Data Visualization with Python – IBM